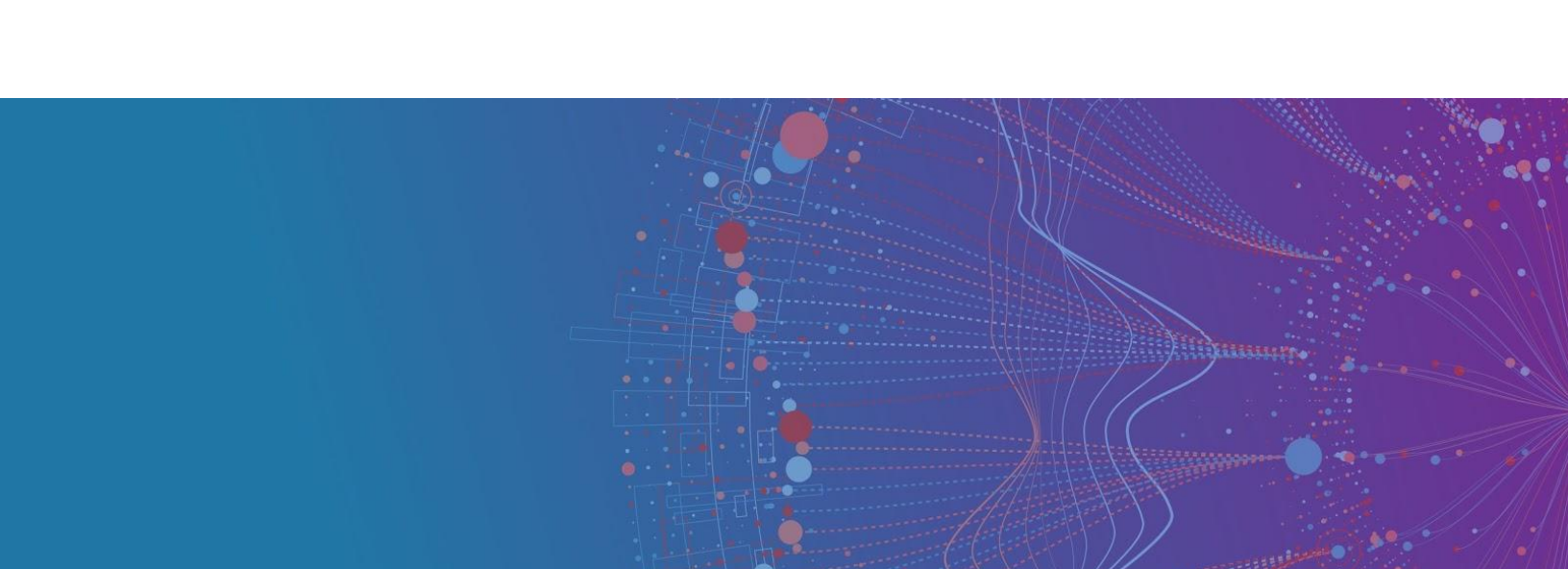




Machine Learning Engineering & AI Bootcamp Capstone

Step 11: Deploy Your Application to Production

Summary

Time Estimate: 15 - 20 hours

Now that you've refactored your code and designed an architecture for your system, it's time to actually implement the architecture and deploy it to production.

The exact steps you follow here will depend on your project. You and your mentor will work on that together. Here are a few common steps that MLEs might follow:

1. **Step One:** Implement your data pipeline. This is where all of the engineering skills you've learned will come in handy.
 - a. Implement the right systems for storing your data, loading it for training your model, and making predictions with your model.
 - b. Make sure you log necessary and sufficient information for monitoring and debugging your system.

- c. Test each component and subsystem as much as possible as you build it out. It's much easier to find bugs in smaller systems than larger ones
- 2. **Step Two:** Design an API using a tool such as Swagger. If you choose to use a managed ML service (e.g. Domino, AWS SageMaker, Google Cloud ML), they might provide out-of-the-box API calls. That's completely fine!
 - a. As you design your API, try to make sure your API follows [good design principles](#).
 - b. Use a tool such as [Flask](#) to create a simple user interface for your API
- 3. **Step Three:** Test your API using tools such as [Postman](#) or [Swagger](#). Make sure you test both read and write functions and various corner cases.
- 4. **Step Four:** Document your API. Using a tool such as [Sphinx](#) lets you automatically create beautiful API documentation from annotations in your code. Add your documentation to your GitHub repository.
- 5. **Step Five:** Add instructions at the very beginning of the README for your GitHub repository on how to install and build your project from scratch, as well as how to access and use your application.
- 6. **Step Six:** Include a Jupyter notebook in the repo that demonstrates how to use the API, and why it is useful. Writing a blog-style article is even better. Look at [fast.ai](#) to see how they do it.
- 7. **Step Seven:** Submit a link to your GitHub repository.